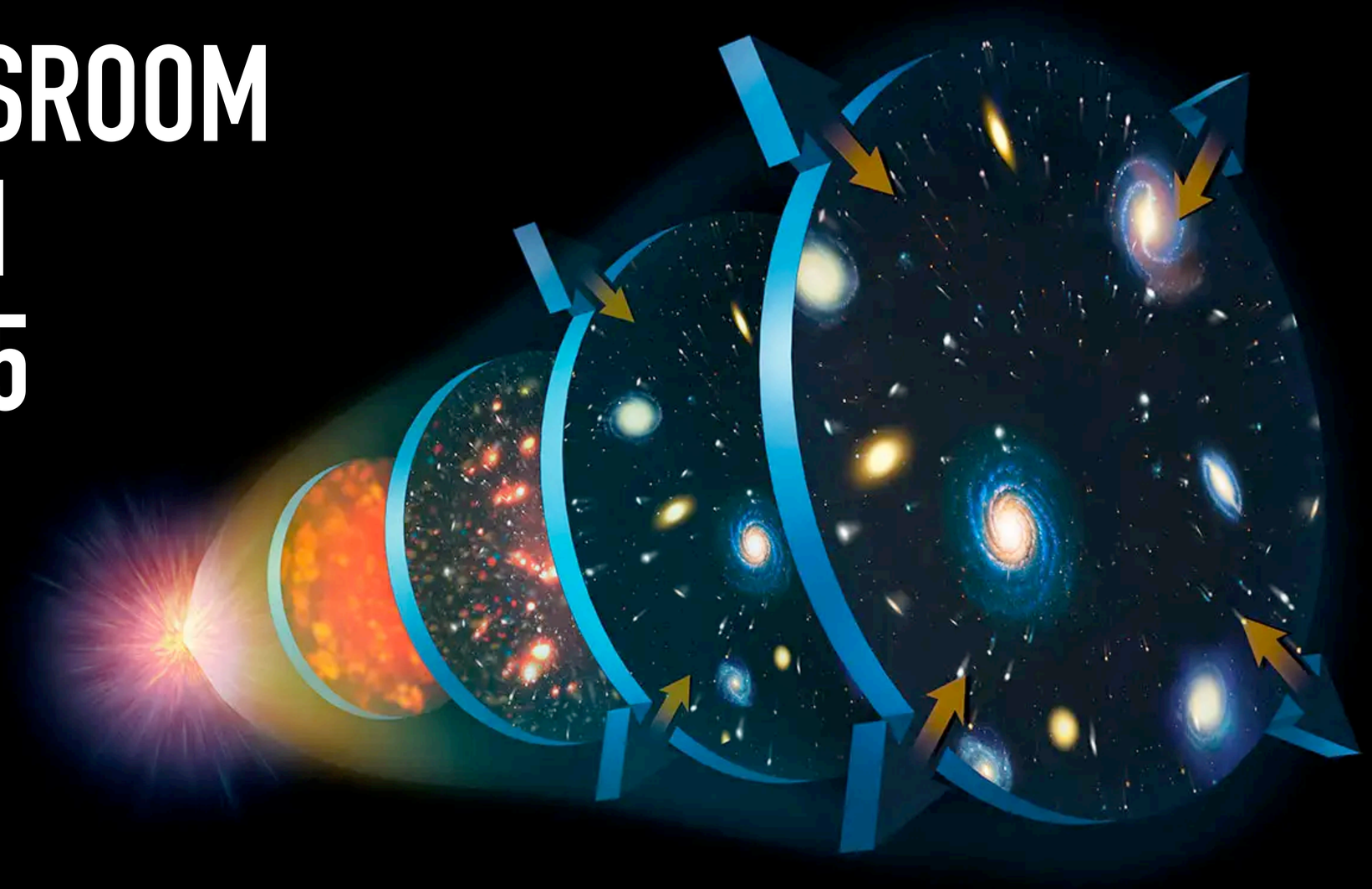
UNIVERSITY OF BIRMINGHAM | GRAVITATIONAL WAVE ASTRONOMY

# OBSERVATIONAL COSMOLOGY

PATRICIA SCHMIDT

P.SCHMIDT@BHAM.AC.UK

# FLIPPED CLASSROOM SESSION 9.10.2025



# QUESTIONS FOR TODAY'S SESSION

What is the cosmic distance ladder? Describe each "rung".

What are standard candles? Give a definition and describe the different types.

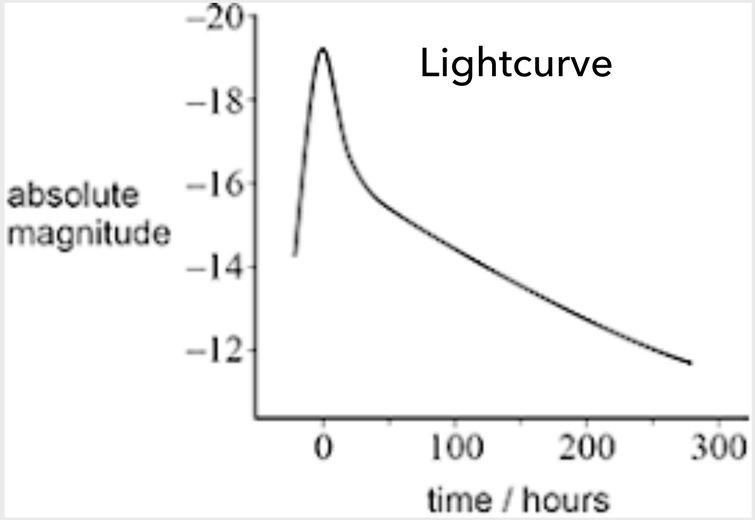
# STANDARD CANDLES

- Standard candles are astronomical objects with **known intrinsic luminosity**
- Discovered by Henriette Swan Leavitt by studying variable stars in the Magellanic Clouds

- **RR Lyrae** variable stars
- **Cepheids:** Variable stars with well-defined pulsation period and amplitude
  - Determine the true luminosity L (magnitude M) by measuring the pulsation period of Cepheids in the SMC (Leavitt's law): $M_\lambda = a_\lambda \log(P) + b_\lambda$
  - Determine the distance by relating the apparent magnitude m to the absolute magnitude M

$$m - M = 5\log_{10}\left(\frac{d}{1\mathrm{Mpc}}\right) + 25 = 5\log_{10}\left(\frac{d}{1\mathrm{pc}}\right) - 5$$

**"Distance modulus"**




- **Type 1a supernovae:** Occur in binary systems where one star is a white dwarf
  - White dwarfs have a common critical mass at which they explode
  - Consistent peak luminosity due to critical mass threshold
  - Peak luminosity tightly coupled to shape of light curve (rise & fall)

# STANDARD CANDLES

- Standard candles need to be **calibrated**.

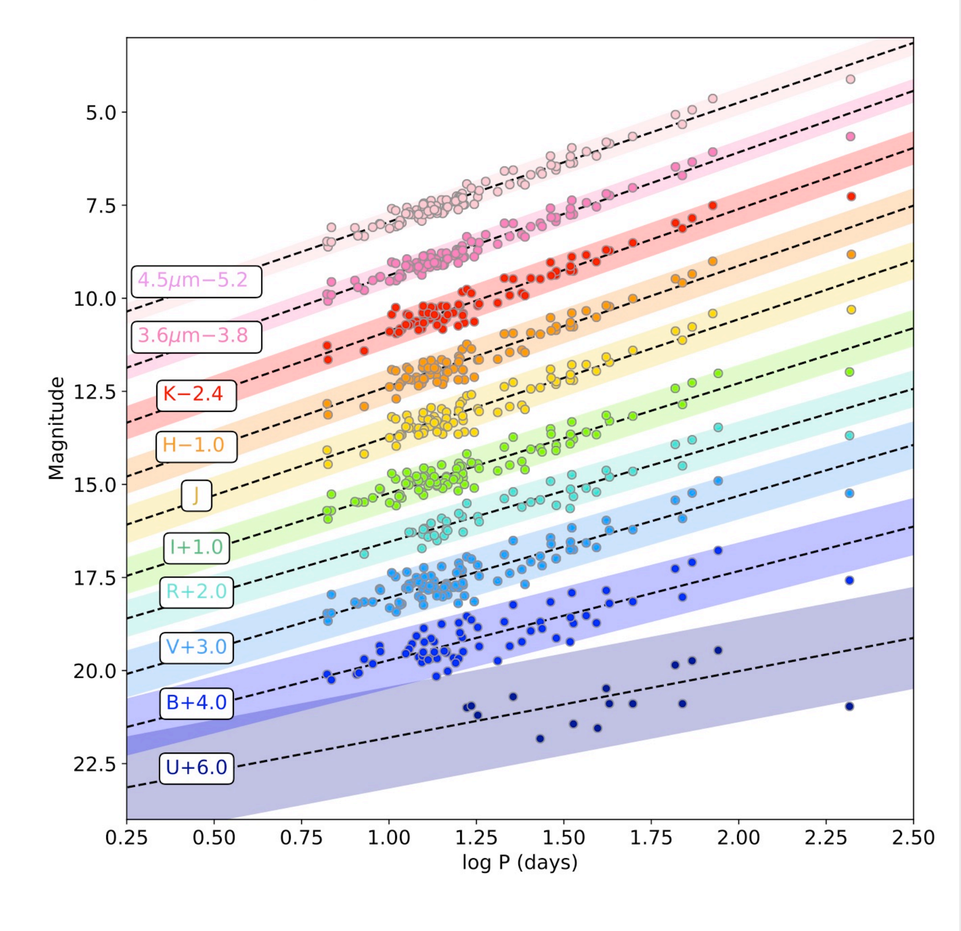
- Examples:
  - Measuring the flux of two Cepheids with the same period in two different galaxies only gives the <u>relative distance</u>:

$$\frac{F(C_1)}{F(C_2)} = \frac{L}{4\pi d(C_1)^2}\frac{4\pi d(C_2)^2}{L} \quad \Rightarrow \quad \frac{d(C_1)}{d(C_2)} = \sqrt{\frac{F(C_2)}{F(C_1)}}$$

  - To determine the <u>absolute distance</u> to a Cepheid, another way of determining distances needs to be used, e.g. the parallax method for nearby Cepheids.
  - These measurements are then used to normalise the period-luminosity relationship.

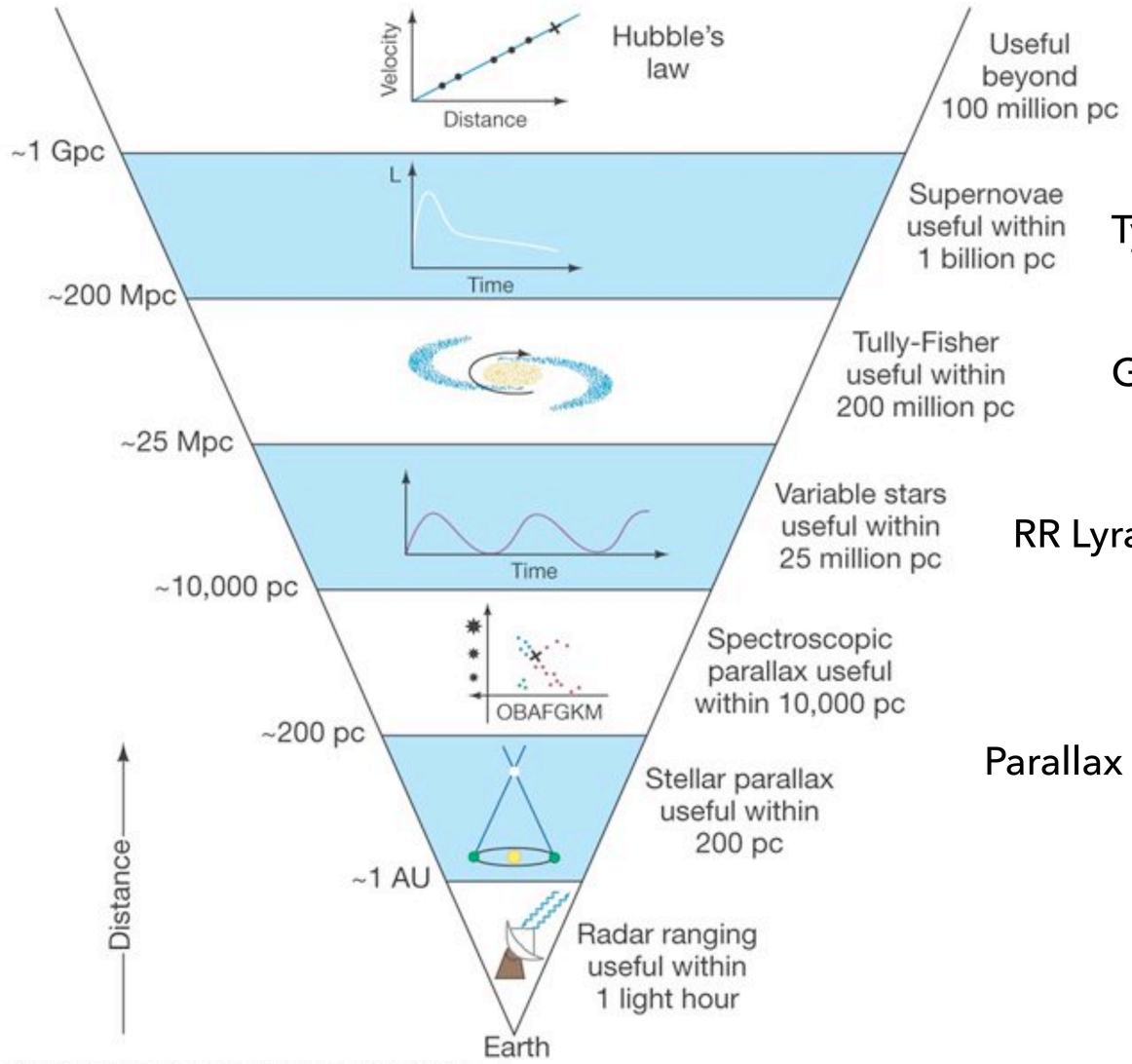
- Fun fact: The Hubble Space Telescope (HST) was built to measure $H_0$ with Cepheids

Cepheids in the SMC:




[Scowcroft+, 2016]

# The "Cosmic Distance Ladder"

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Type 1a supernovae

Galaxy rotation curves

RR Lyrae and Cepheids standard candles

Parallax measurements

# THE COSMIC DISTANCE LADDER

- Galaxy rotation curves (require calibration)
  - Tully-Fisher relation for spiral galaxy & Faber-Jackson relation for elliptical galaxies
    - Relate the intrinsic luminosity of the galaxy to the dispersion velocity of stars

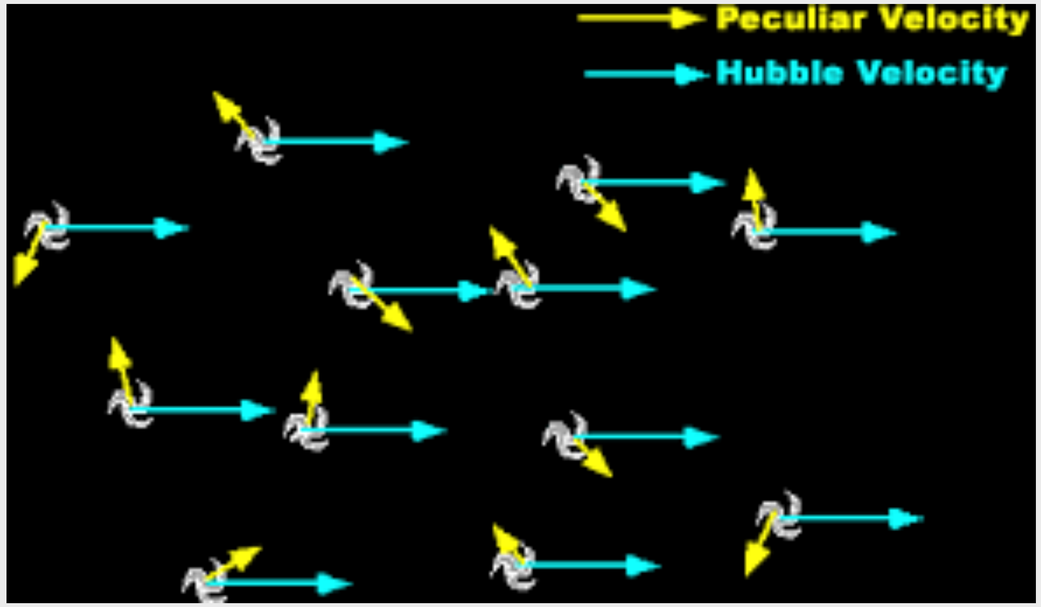
- Some other methods:
  - Main sequence fitting
  - Tip of the red giant branch (TRGB)
  - Luminosity functions (nebulae, globular clusters)
  - Sigma-D relation
  - …

- Gravitational waves = “standard sirens”
  - Allow for a direct measurement of the (luminosity) distance

# HUBBLE'S LAW – REMARKS

- Hubble's law is not exact; most notably it does not hold for nearby galaxies

- The total radial velocity (i.e. along the line-of-sight) of nearby galaxies is dominated by their random motion due to local gravitational interactions (peculiar velocity):

$$v_{\rm total} = v_{\rm pec} + v_H$$

- Only valid for distant galaxies, for which their recessional velocity dominates their total velocity:
  - Galaxies in the Hubble flow